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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/708,932	03/30/2022	Elham RAFIE BORUJENY	93041.3-psw	3872
22828	7590	03/16/2023	EXAMINER	
BENNETT JONES LLP			TSCHEN, FRANCISCO W	
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary

Application No.

17/708,932

Applicant(s)

RAFIE BORUJENY, Elham

Examiner

FRANCISCO W TSCHEN

Art Unit

1712

AIA (FITF) Status

Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 1/10/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1-14 and 17-19 is/are rejected.

8) ☒ Claim(s) 15-16 is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 3/30/2022 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

Applicant's election without traverse of Group I, Claims 1-19 in the reply filed on January 10, 2023 is acknowledged. Claim 20 is now cancelled.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claim(s) 1-4, 6, 8, 17-19 are rejected under 35 U.S.C. 103 as being unpatentable over Liu et al. (b-Ga₂O₃ thin films on sapphire pre-seeded by homo-self-templated buffer layer for solar-blind UV

photodetector, Optical Materials 51, 2016, pp.203-207; Liu) in view of Donmez et al. (Low temperature deposition of Ga₂O₃ thin films using trimethylgallium and oxygen plasma, J.Vac. Sci. Technol. A, 31, 2013, pp.1-4; Donmez)

Re Claims 1, 3, 4, 8: Liu discloses a method for forming β -Ga₂O₃ thin films. Title. Liu forms the films on c-plane sapphire substrates by plasma-assisted molecular beam epitaxy (MBE). Section 2. Experimental; col. 2; p. 203. Liu forms the β -Ga₂O₃ layer by using evaporate gallium with mono-atomic oxygen. Id. The first two monolayers act as a β -Ga₂O₃ homo-self templated buffer layer.

Liu forms a β -Ga₂O₃ layer over the buffer layer. Section 2. Experimental; col.1, p.204, first full para. Liu discloses that the β -Ga₂O₃ layer with the buffer layer achieve better crystalline quality. Section 3. Results and discussion, col.1, p.204, second para.; see also Section 4 Conclusions. p.207.

Liu discloses forming the layer utilizing gallium but does not disclose that the β -Ga₂O₃ is formed by utilizing gallium precursors.

However, Donmez in the same field of gallium oxide films discloses an alternate formation method utilizing plasma enhanced atomic layer deposition (PEALD). Section II. Experiment, 1st para., p.1. Donmez discloses the method wherein the β -Ga₂O₃ is formed. Abstract. The layer is formed utilizing trimethylgallium (TMG) as the gallium precursor and O₂ plasma as the oxygen precursor, and annealed to obtain the desired crystalline phase. Id.

It would have been obvious to someone of ordinary skill in the art at the time the invention was made to modify the method of Liu and form the buffer layer, or the overcoated β -Ga₂O₃ layer, or both layers of Liu by utilizing the method of Donmez because it is known as a suitable way to form the layer with reasonable expectation of success, provide the layer-by-layer control over film growth, and achieve the desired properties including refractive index and roughness.

A finding that one of ordinary skill in the art could have substituted one known element for another, and the results of the substitution would have yielded predictable results supports a prima facie case of obviousness. See MPEP 2143, "KSR" rationale B.

Re Claim 2, 6: Liu discloses the layer is subjected to growth parameter optimization to produce the desired growth rate and thickness. Section 2, Experimental, col.1, p.204. It would have been obvious to someone of ordinary skill in the art at the time the invention was made to modify the method of Liu and optimize the conditions to achieve the desired growth rate and thickness for each layer for the particular use of the product produced.

Claim(s) 5-6 are rejected under 35 U.S.C. 103 as being unpatentable over Liu et al. (Liu) in view of Donmez et al. (Donmez), and Bakke et al. (Nanoengineering and interfacial engineering of photovoltaics by atomic layer deposition, Nanoscale, 2011, 3, pp.3482–3508; Bakke)

Liu discloses the use of gallium oxide layers in the field of optoelectronics. See Introduction. Liu does not explicitly disclose depositing just a monolayer of the gallium oxide.

However, Bakke in the same field of optoelectronics (photovoltaics) discloses the use of buffer layers formed via atomic layer deposition. Abs. Bakke discloses that with ALD the reaction is limited to a monolayer per cycle, col. 2, p. 3484, where the process is repeated until the desired thickness is achieved. Id. The ALD process is known in the art to form buffer layer. para 1, col. 2, p. 3485, Bakke discloses that the ALD is gaining more acceptance because it allows selecting different materials and compositions for the buffer layer and vary its properties. para 1. col.2, pp.3488 (“spawned more interest in the ALD community thus far because the materials and composition of the buffer layer”). Finally, Bakke discloses that ALD is useful because it helps control the thickness of the composition because the thickness has an effect on the buffer layer and its performance. para 1, col.1, p. 3490.

The determination of optimum or workable ranges of the buffer layer thickness to achieve advantages of the buffer layer would have been characterized by routine experimentation. See MPEP 2144.05 IIB

It would have been obvious to someone of ordinary skill in the art at the time the invention was made to modify the method of Liu and optimize the buffer layer thickness through the use of ALD as taught by Bakke. Bakke discloses that the buffer layer thickness has an effect on the properties of the buffer layer and its performance. Liu/Donmez disclose that the buffer has alignment and interaction with the substrate and overlying layers. Thus, the combination allows formation of the layer in a layer-by-layer fashion, including forming a monolayer of the composition to provide the properties desired without waste of chemical precursors and time.

Claim(s) 7 and 9 are rejected under 35 U.S.C. 103 as being unpatentable over Liu et al. (Liu) in view of Donmez et al. (Donmez), and Huang et al. (On the response of gamma irradiation on atomic layer deposition-grown β -Ga₂O₃ films and Au- β -Ga₂O₃ -Au deep ultraviolet solarblind photodetectors, J. Vac. Sci. Technol. A 38, 062409 (2020), pp.1-6; Huang)

Re Claim 7: Liu/Donmez disclose the method as shown above including forming the layer of step b) via atomic layer deposition. The combined references do not disclose utilizing triethylgallium gas as the gallium precursor.

However, Huang is in the same field of forming β -Ga₂O₃ films. Title. Huang discloses forming the films via atomic layer deposition, which is the same method of Donmez. Section II Experiment. Huang discloses utilizing different precursors to form the layer. Id. The gallium precursor is selected from triethylgallium (TEG). Id. The oxygen precursor is selected from nitrous oxide. Id. The use of these precursors results in higher overpressure of oxygen radicals than using just oxygen or water at the same deposition temperature. first full para. col.1, p. 2.

It would have been obvious to someone of ordinary skill in the art at the time the invention was made to modify the method of Liu/Donmez and form the beta-gallium oxide film with the precursors of Huang. Both Huang and Donmez disclose using atomic layer deposition is suitable for the formation of the layers. Huang utilizes different precursors with result in advantage as it relates to overpressure of oxygen radicals to aid in forming of the film without defective structures.

Furthermore, a finding that one of ordinary skill in the art could have substituted one known element for another, and the results of the substitution would have yielded predictable results supports a prima facie case of obviousness. See MPEP 2143, "KSR" rationale B.

Re Claim 9: Liu/Donmez/Huang disclose the atomic layer deposition process to form the layer as shown above. Atomic deposition by definition is a deposition technique based on the sequential use of gas phase chemical process. Thus it is understood by the ordinary artisan that the method of Huang comprises pulsed doses of TEG and Nitrous oxides. Donmez discloses pulsing the gallium precursor for 0.015s and the oxygen precursor for 2-60 seconds. 1st para., Experiment Section.

In the absence of the timing, it would have been obvious to someone of ordinary skill in the art at the time the invention was made to look at timing of the prior art and modify the method of Liu/Donmez/Huang and pulse the precursors in a similar fashion as Donmez because it is known to provide the right amount of reactive radicals to form the layer via layer-by-layer fashion.

However, Examiner notes that the pulsing time changes the amount of the precursors, and basically a measure of concentration of the precursor within the system. The courts have held that differences in concentration or temperature will not support the patentability of subject matter encompassed by the prior art unless there is evidence indicating such concentration or temperature is critical. See MPEP 2144.05 II A.

Claim(s) 10-11 are rejected under 35 U.S.C. 103 as being unpatentable over Liu et al. (Liu) in view of Donmez et al. (Donmez), Yao et al. ((Invited) Growth and Characterization of α -, β -, and ϵ -Ga₂O₃ Epitaxial Layers on Sapphire, ECS Trans, 80, 7, 2017, pp. 191-196; Yao), and Nakayama et al. (US 2004/0188693; Nakayama). Wei (Structural properties of b-Ga₂O₃ formed by dry thermal oxidization process on GaN, Material Science in Semiconductor Processing, 15, 2012, pp.578-581; Wei) is presented as evidence of the crystalline nature of the gallium oxide.

Liu/Donmez teach the method as shown above but do not disclose the buffer layer comprises alpha-Ga₂O₃.

However, Yao in the same field of growing gallium oxide film on sapphire discloses forming alpha, beta and epsilon gallium oxide. Title. Yao discloses that an alpha-Ga₂O₃ is formed on the sapphire layer. Figure 3. The alpha-Ga₂O₃ appears to serve as a better buffer with the sapphire because the mismatch between the alpha phase is less than between beta or epsilon phases. 1st para., p.194. (it is noted buffers improved crystallographic alignment, due to less mismatch would result in better inhibition of cracks as known in the art; and more uniformity of film growth).

It would have been obvious to someone of ordinary skill in the art at the time the invention was made to modify the method of Liu/Donmez and utilize alpha-Ga₂O₃ for the buffer phase because it has a less mismatch with sapphire than the beta phase which would be an advantage to form the subsequent beta layer as the buffer layer improves the subsequent deposition as disclosed by Liu.

Liu/Donmez/Yao teach the method as shown above, but does not disclose that the buffer layer is formed via a GaN layer.

However, Nakayama discloses a method of forming a GaN layers on a sapphire substrate. [0076]. The GaN layer is subjected to oxidation to form an oxide layer of Ga₂O₃. [0094]. The oxidation is

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done at a range of temperature at 1000C. [0092]. This would have resulted in the alpha gallium phase as evidenced by Wei based on processing at 1000C. See Abs.

It would have been obvious to someone of ordinary skill in the art at the time the invention was made to modify the method of Liu/Donmez and form the buffer layer including alpha- Ga_2O_3 since it is known as a suitable method to form the Ga_2O_3 over a sapphire substrate to produce the desired semiconductor application which Donmez contemplates.

Claim(s) 12-14 are rejected under 35 U.S.C. 103 as being unpatentable over Liu et al. (Liu) in view of Donmez et al. (Donmez), Yao et al. (Yao), Nakayama et al. (Nakayama), and Motamedi (A route to low temperature growth of single crystal GaN on sapphire, J. Mater. Chem. C, 3, 2015, 7428-7436; Motadeni) . Wei (Wei) is presented as evidence of the crystalline nature of the gallium oxide.

Liu/Donmez/Yao/Nakayama disclose the method as shown above. The method forms a GaN layer and converts it to a Ga_2O_3 layer on a sapphire substrate. However, it does not disclose utilizing TEG and nitrogen precursor to form the layer.

However, Motadeni discloses forming GaN on sapphire. Title. Motadeni forms the method utilizing ALD by cycling four consecutive pulses, 0.02s TEG, 7s Argon purge, 15s N_2/H_2 , and 5s Argon purge repeated for several cycles to obtain the desired layer thickness. Section 2. Experiments, col.1.

It would have been obvious to someone of ordinary skill in the art at the time the invention was made to modify the method of Liu and form the GaN layer by the method of Motadeni because it is known as a suitable way to form the GaN layer at low temperatures having desired properties and direction. Abs. Furthermore forming the layers via ALD would be compatible with the method of Donmez which also discloses ALD.

Allowable Subject Matter

Claims 15 and 16 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

- a. Oda et al. (Crack-free thick ($>5\text{ }\mu\text{m}$) $\alpha\text{-Ga}_2\text{O}_3$ films on sapphire substrates with $\alpha\text{-(Al,Ga)}_2\text{O}_3$ buffer layers, Jpn. J. Appl. Phys. 55, 1202B4 (2016)) discusses the formation of buffer layers to prevent crack formation. See Introduction.
- b. Biyikli et al. Atomic layer deposition: an enabling technology for the growth of functional nanoscale semiconductors. Semicond. Sci. Technol., 32, 2017 093002. Discloses the state of the art for Atomic Layer Deposition including formation of GaN and Ga_2O_3 layers.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to FRANCISCO W TSCHEN whose telephone number is (571)270-3824. The examiner can normally be reached M-Th: 9-4, F:9-8 (PHP).

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Michael Cleveland can be reached on 571-272-1418. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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FRANCISCO TSCHEN
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